MILITARY SPECIFICATION
BEACON SET, RADIO AN/TRN-30( )
(UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of radio beacon set which operates on frequencies in the ranges of 200 to 535 kHz and 1605 to 1750 kHz and is designated as Beacon Set, Radio AN/TRN-30( ) (See 6.1 and 6.5).

Transmitter, Radio T-1199( )/TRN-30
Amplifier-Coupler AM-6417( )/TRN-30
Antenna AS-2633( )/TRN-30
Antenna AS-2634( )/TRN-30
Case, Transmitter CY-7142( )/TRN-30
Case, Amplifier-Coupler CY-7143( )/TRN-30

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS:

FEDERAL

PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed A and Lock-corner
PPP-F-320	Fiberboard, corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes
PPP-T-76	Tape, Pressure-sensitive Adhesive paper, Water Resistant, (For Carton Sealing)
QQ-S-781	Steel Strapping; Flat

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-713	Twine, Impregnated, Lacing and Typing
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-P-55110	Printed Wiring Boards

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-669	Loading Environment and Related Requirements for Platform Rigged Airdrop Material
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28 volt DC Transient Characteristics for Military Vehicles

DRAWINGS

USAECOM

DL-SM-A-550412	Transmitter, Radio T-1199( )/TRN-30
DL-SM-A-550749	Amplifier-Coupler AM-6417( )/TRN-30
DL-SM-A-551037	Antenna AS-2633( )/TRN-30
DL-SM-A-551038	Antenna AS-2634( )/TRN-30
DL-SM-A-550392	Case, Transmitter CY-7142( )/TRN-30
DL-SM-A-550393	Case, Amplifier-Coupler CY-7143( )/TRN-30
SC-A-678026	Burn-In/Debugging Requirements for Beacon Set, Radio AN/TRN-30( )(Units Of)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Construction. - Construction shall be in accordance with the following drawing and data lists:

DL-SM-A-550412	Transmitter, Radio T-1199( )/TRN-30
DL-SM-A-550749	Amplifier-Coupler AM-6417( )/TRN-30

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DL-SM-A-551037	Antenna AS-2633( )/TRN-30
DL-SM-A-551038	Antenna AS-2634( )/TRN-30
DL-SM-A-550392	Case, Transmitter CY-7142( )/TRN-30
DL-SM-A-550393	Case, Amplifier-Coupler CY-7143( )/TRN-30

3.2 Preproduction samples. - The contractor shall furnish the specified number of preproduction samples on order for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2)

3.3 Cleaning. -

3.3.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Post cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, and other parts.

3.3.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.5 Finish, protective (See 4.4). - The units shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be Film M in accordance with MIL-F-14072.

3.6 Securing of parts. -

3.6.1 General. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

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3.7 Marking (See 4.4). -

3.7.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be engraved in accordance with Group 1 of the specification.

3.7.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum dissasembly of the unit(s).

3.7.3 Serial numbers. - The unit(s) listed in paragraph 1.1 which are being procured on this contract shall be serial numbered.

3.8 Soldering. -

3.8.1 Soldering. - Soldering shall conform to Requirement 5 of MIL-STD-454 and the equipment drawings.

3.8.2 Rosin-base residues. - Flux residues shall be removed by applying organic solvents. Mechanical means such as agitation, brushing, etc., may be used in conjunction with the solvents. The cleaning solvents and methods used shall not have any deleterious effect on the parts, materials, and assemblies being cleaned.

3.8.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the unit(s); flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.9 Wiring, cabling, bonds and grounds. -

3.9.1 Cabling. - Insulated wires shall be formed into cables except where operation of the unit(s) would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

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3.9.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.9.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.9.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (1) during and after subjection of the unit(s) to specified service conditions or (2) after service or repair of the unit(s) in a normal manner.

3.9.5 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.9.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.9.7 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.9.8 Bonds and grounds. - Mating surfaces requiring RF gaskets and internal grounds and bonds required to reduce current path return, RF potentials and shock hazards shall be free of anodic film, non-conductive grease, paint or lacquer.

3.10 Semiconductors (transistors & diodes) (See 4.4). - These electron devices (semiconductors and diodes) shall conform to Specification MIL-S-19500, and the equipment drawings.

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3.11 Tropicalization of materials. - Materials shall be treated as follows:

3.11.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the materials or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.11.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment. or during use of the finished items when used for the purpose intended.

3.11.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11.4 Statement of treatment. - The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.12 Service Conditions (See 4.8). - The equipment shall meet the following service conditions.

3.12.1 Temperature (See 4.8.1). -

3.12.1.1 High Temperature. - The equipment shall meet full performance requirements after storage at 155°F and when operated at 155°F.

3.12.1.2 Low Temperature. - The equipment shall meet full performance requirements after storage at -65°F and when operated at -50°F. The time duration of low temperature storage at -65°F shall be 2 hours.

3.12.2 Humidity (See 4.8.2). - The equipment shall meet full performance requirements when subjected to high relative humidities of 100% at all temperature below +85°F and relative humidities corresponding to 85°F dew point at all temperatures between +85°F and 120°F and low relative humidities of 5% at temperature of +120°F.

3.12.3 Altitude (See 4.8.3). - Operating up to 10,000 feet above sea level, non-operating up to 50,000 feet above sea level.

3.12.4 Vibration (See 4.8.4). -

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3.12.4.1 Resonance search (See 4.8.4.1). - With the equipment shock mounts (if any) blocked, the amplitude of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.

3.12.4.2 Bounce (Loose Cargo) (See 4.8.4.2). - The equipment in its transit case, or cases, shall be capable of withstanding the shocks normally induced during field transport as loose cargo.

3.12.5 Shock (See 4.8.5). -

3.12.5.1 Bench handling (See 4.8.5.1). - The equipment shall be capable of withstanding shock encountered in servicing.

3.12.5.2 Drop (See 4.8.5.2). - The equipment in its transit case(s) shall be capable of withstanding shocks induced by rough handling and loading of equipments.

3.12.6 Dust (See 4.8.6). - The equipment in its physical operating configuration shall be capable of withstanding dust particles as encountered in coastal and desert areas without penetration.

3.12.7 Immersion (See 4.8.7). - The equipment in its transit cases CY-7142 or CY-7143 shall be capable of being immersed to a covering depth of 3 feet of fresh water for a period of 2 hours without evidence of any water leakage.

3.12.8 Rain (See 4.8.8). - The equipment in its operating configuration shall withstand rain for the required durations without leakage. During each 30 minute rain period, the wind shall be maintained for 15 minutes commencing 5 minutes after the start of the period specified in the procedure. The equipment shall be operated during the last 10 minutes of the 30 minute period.

3.12.9 Salt Fog (See 4.8.9). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.12.10 Fungus (See 4.8.10). - The unit(s) shall show no evidence of viable fungus after being subjected to 28 days of fungus test.

3.12.11 Air delivery (See 4.8.11). - The equipment shall be capable of air delivery by parachute at a maximum rate of descent of 25 feet per second, and shall withstand an air drop test as specified.

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3.12.12 Reliability (See 4.8.12). - Each unit shall demonstrate a reliability sufficient to provide mean-time-between failures (MTBF's) as follows:

a. Transmitter T-1199( )/TRN-30(V) - 1200 hr MTBF, assuming an exponential distribution.

b. Transmitter T-1199( )/TRN-30(V) and Amplifier-Coupler AM-6417( )/TRN-30(V) operating in Semi-Fixed Mode - 1000 hr MTBF, assuming an exponential distribution.

c. Ambient operation temperature environment +125oF.

d. Operation-Continuous

e. Ground Environment.

3.13 Electromagnetic Compatibility (See 4.9). - The Transmitter/Amplifier-Coupler(s) operating in all modes (CW and MCW) shall meet requirements CE 01, CE 03, CS 01, CS 02, CS 06, RE 02, RE 03, and RS 03 of MIL-STD-461 as Class 1B equipment, except as indicated herein. For bonds and grounds, see 3.9.8.

3.13.1 Conducted emission CE 01 (See 4.9.1). - The equipment(s) shall meet the narrowband requirements of Method CE 01.

3.13.2 Radiated emission RE 02 (See 4.9.2). - In an open area under standard test conditions of 4.7 equipment(s) shall meet the requirements of RE02 over the frequency range of 14kHz to 1.0 GHz. The requirements shall be met with the equipment(s) operating under standard test conditions of 4.7. During production testing RE 02 shall be performed within a shielded inclosure at a 1.0 meter test distance and the limits shall be relaxed 30 dB.

3.13.3 Spurious and harmonic emissions RE 03 (See 4.9.3). - In an open area utilizing the equipment antenna(s) with radial ground planes and guy wires, the equipment(s) shall meet the requirement of RE 03 with the equipment operating into its antenna(s). When operating in the semi-fixed mode, the second harmonic shall be suppressed a minimum of 68 dB in the CW mode and a minimum of 65 dB in the MCW mode and all other harmonic and spurious outputs shall be within the requirements of RE 03. The harmonic and spurious suppression in terms of dB down from the fundamental shall be calculated from Figure 16 of MIL-STD-461.

3.13.4 Radiated susceptibility RS 03 (See 4.9.4). - The equipment(s) shall meet the requirements of RS 03 over the frequency range of 14 kHz to 1.0 GHz. These requirements shall be met utilizing a shielded inclosure with the equipment operating under standard test conditions of 4.7.

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3.14 Electrical requirements, Transmitter T-1199( )/TRN-30 (See 4.10). - Unless otherwise specified, the transmitter shall meet all the following requirements (3.14.1 thru 3.14.11 under Standard Test Conditions after 30 seconds of operation following completion of the tuning cycle.

3.14.1 Transmitter, operating (See 4.10.1). - The unit shall automatically tune to all frequencies in the range of 200.0 to 535.0 kHz and 1605.0 to 1750.0 kHz in 0.5 kHz steps as selected with input voltages from 22 to 30 VDC.

3.14.2 Transmitter lockout (See 4.10.2). - When operating, the unit shall automatically prevent any frequency selected outside the 199.5 to 536.0 kHz and 1604.5 to 1750.5 kHz bands from being generated.

3.14.3 Exciter, operating (See 4.10.3). - With the transmitter operating as an exciter for the Amplifier-Coupler, it shall automatically tune to any frequency selected in the range of 200.0 to 535.0 kHz.

3.14.4 Exciter, lockout (See 4.10.4). - With the transmitter operating as an exciter for the Amplifier-Coupler, it shall automatically prevent any frequency selected outside the 199.5 to 536.0 kHz band from being generated.

3.14.5 Frequency stability (See 4.10.5). - With the transmitter operating in its frequency range, the measured frequency shall be within 0.0001 percent of its indicated frequency with input voltages from 22 to 30 VDC.

3.14.6 Beacon identification (See 4.10.6). -

3.14.6.1 Code characters (See 4.10.6.1). - The transmitter shall provide selection of four letter Morse Code groups. Each of the four characters are letters of the alphabet or blank characters. Inclusion of blank characters provides the capability of generating one, two, or three letter groups as well as the four letter groups.

3.14.6.2 Code rate (See 4.10.6.2). - The code rate shall be variable from seven to twenty words per minute. This shall be accomplished by varying the dot length from less than 50 milliseconds to greater than 150 milliseconds. The dash to dot ratio shall be fixed at 3 to 1.

3.14.6.3 Word spacing (See 4.10.6.3). - Selected code groups shall be repeated at 3.5 $\pm$ 1.0 second intervals.

3.14.6.4 Modulation frequency (See 4.10.6.4). - The frequency of the amplitude modulation shall be 1020 $\pm$ 10 Hz.

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3.14.6.5 Percent modulation (See 4.10.6.5). - The transmitter shall provide the following minimum percent modulation of the radiated carrier:

Frequency(kHz)	Minimum Percent Modulation
200	50
260	60
310	70
360	75
535	80
1605	85

3.14.7 Input power (See 4.10.7). -

3.14.7.1 Power, steady state (See 4.10.7.1). - The transmitter shall give its specified performance when operated on steady state voltages between 22 and 30 VDC.

3.14.7.2 Over voltage transient (See 4.10.7.2). - The transmitter shall not sustain damage and shall give its specified performance when subjected to the 40 volt over voltage transient of MIL-STD-1275.

3.14.7.3 Under voltage transient (See 4.10.7.3). - The transmitter shall not sustain damage when subjected to the 15 volt under voltage transient of MIL-STD-1275. It shall also automatically resume specified performance upon removal of the transient voltage.

3.14.7.4 Power consumption (See 4.10.7.4). - The transmitter input current shall not exceed 3.3 amperes at normal input voltage.

3.14.8 Antenna tuning (See 4.10.8). - The maximum tune time for the transmitter shall be not more than 2.0 minutes when operating as specified.

3.14.9 Output signals (See 4.10.9). -

3.14.9.1 Transmitter, RF (See 4.10.9.1). - With the transmitter operating as specified with input voltages from 22 to 30 VDC, the RF current shall be as indicated:

Frequency (kHz)	Simulated Load	Minimum RF Current (Amps)
200 - 360	30 Foot Antenna	0.8
360 - 535	30 Foot Antenna	1.0
1605 - 1750	30 Foot Antenna	1.1
1605 - 1750	15 Foot Antenna	1.0

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3.14.9.2 Exciter (See 4.10.9.2). -

3.14.9.2.1 Exciter, RF (See 4.10.9.2.1). - With the transmitter operating as an exciter for the Amplifier-Coupler, the RF signal present at the transmitter RF connector, when terminated with 50 ± 5 ohm load shall be as follows:

Waveform:	Square Wave
Rise Time:	50 ± 20 nano seconds
Symmetry:	± 1.0% maximum
Power:	4.0 ± 1.0 watts

3.14.9.2.2 Exciter, Audio (See 4.10.9.2.2). - With the transmitter operating as an exciter for the Amplifier-Coupler, the audio signal present at the transmitter audio connector shall have an audio frequency of 1020 Hz ± 10 Hz and a power level of 0.0 ± 1.0 dBm measured into a 600 ohm load and shall be balanced to ground.

3.14.9.2.3 Exciter, logic (See 4.10.9.2.3). - With the transmitter operating as an exciter for the Amplifier-Coupler, the signals which provide control filter selection in the Amplifier-Coupler shall have the following characteristics:

Frequency (kHz)	Filter Logic (NOTE: Pin Connections of 1J2)			
	C	P	N	L
200.0	High	Low	Low	Low
285.0	Low	High	Low	Low
325.0	Low	Low	High	Low
535.0	Low	Low	Low	High

High is defined as +2.5 to +5.0 VDC.
Low is defined as 0.0 to +0.4 VDC.

3.14.10 Meter, voltage (See 4.10.10). - The voltage meter shall indicate in the green area when the input voltages from 22.0 to 30.0 VDC are applied to the power input terminal.

3.14.11 Meter, RF (See 4.10.11). - The transmitter RF meter shall indicate in the green area when antenna tuning is accomplished and the RF output power level is as listed in 3.14.9.1.

3.15 Electrical requirements, Amplifier-Coupler AM-6417( )/TRN-30 (See 4.11). - Unless otherwise specified, the amplifier-coupler shall meet all the following requirements (3.15.1 thru 3.15.6) under Standard Test Conditions after 30 seconds of operation following completion of tuning cycle.

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3.15.1 Input power (See 4.11.1). -

3.15.1.1 Power, steady state (See 4.11.1.1). - The Amplifier-Coupler shall give its specified performance when operated on steady state voltages between 26 and 30 VDC.

3.15.1.2 Over voltage transient (See 4.11.1.2). - The Amplifier-Coupler shall not sustain damage and shall give its specified performance when subjected to the 40 volt over voltage transient of MIL-STD-1275.

3.15.1.3 Under voltage transient (See 4.11.1.3). - The Amplifier-Coupler shall not sustain damage when subjected to the 15 volt under voltage transient of MIL-STD-1275. It shall automatically resume specified performance upon removal of the transient voltage.

3.15.1.4 Power consumption (See 4.11.1.4). - The Amplifier-Coupler input current for the Semi-Fixed mode of operation shall not exceed 30 amperes. In any of the Tactical modes the input current shall not exceed 8.0 amperes.

3.15.2 Percent modulation (See 4.11.2). - When the Amplifier-Coupler is excited by the specified modulated signal, it shall provide the following minimum percent modulation at the RF output terminal:

Frequency (kHz)	Minimum Percent Modulation
200.0	65
260.0	70
310.0	75
360.0	80
535.0	85

3.15.3 Antenna tuning (See 4.11.3). - The maximum tune time for the amplifier-coupler shall be not more than 2.0 minutes when operating as specified.

3.15.4 Output power (See 4.11.4). -

3.15.4.1 Tactical mode, RF (See 4.11.4.1). - With the Amplifier-Coupler operating into the required load and input voltages from 26 to 30 VDC, the RF current shall be as indicated:

Frequency (kHz)	Minimum RF Current (Amps)
200.0 - 259.5	1.2
260.0 - 309-5	1.0
310.0 - 359.5	0.9
360.0 - 535.0	0.8

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3.15.4.2 Semi-Fixed Mode, RF (See 4.11.4.2). - With the Amplifier-Coupler operating into the required load and input voltages from 26 to 30 VDC, the RF current shall be as indicated:

Frequency (kHz)	Minimum RF Current (Amps)
200.0 - 259.5	2.3
260.0 - 309.5	2.6
310.0 - 359.5	2.5
360.0 - 535.0	2.3

3.15.5 Exciter Enable (See 4.11.5). - When the audio cable is connected to the transmitter and Amplifier-Coupler audio connectors, the transmitter antenna tuning and the 1605 to 1750 kHz frequency range shall automatically disable and the RF output shall automatically switch from the transmitter antenna to the RF connector.

3.15.6 Meter, RF (See 4.11.6). - The Amplifier-Coupler RF meter shall indicate in the green area when antenna tuning is accomplished and the RF output level is as listed in 3.15.4.

3.16 Burn-in/debugging (See 4.6). - Each unit on order shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the burn-in/debugging as specified.

3.17 Interchangeability (See 4.13). - Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable. Items which are covered by specified electrical performance requirements shall meet such specified performance requirements prior to installation into the unit(s). After installation of these items and prior to submission of unit(s) for the quality performance specifications, adjustments shall be made, if required, to permit the unit(s) to meet full equipment specification requirements. Alteration of any items, or the unit(s) shall not be permitted.

3.18 Operability (See 4.14). - Each unit(s) shall be subjected to the operational inspection as specified, after applicable processes, inspections and tests are complete, to determine that each unit on order is operable prior to packaging.

3.19 Technical literature and spare parts (See 6.2). - Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the equipment furnished on order.

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3.20 Workmanship (See 4.12). - The equipment shall be manufactured and assembled in accordance with the equipment drawings (See 3.1) and the following requirements and paragraphs herein:

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Soldering	Requirements 5
Workmanship	Requirements 9

Paragraphs herein:

Cleaning	3.3
Controls	3.4
Securing of parts	3.6
Soldering	3.8
Wiring, cabling, bonds and grounds	3.9
Tropicalization of materials	3.11

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents (See 4.4)

c. Quality Conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery (See 4.15 and 4.16).

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4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (See Tables I, II and III respectively). The preproduction inspection will normally be performed in this order:

Sample 1: (1) High Temperature; (2) Low Temperature; (3) Rain; (4) Humidity; (5) Salt Fog; Sample 2: (1) Altitude; (2) Dust; (3) Resonance Search; (4) Shock-Bench Handling; Sample 3: (1) Fungus; Sample 4: (1) EMC; (2) Shock-Drop; (3) Bounce (loose cargo); (4) Air delivery; (5) Immersion; other preproduction inspections may precede, follow or be interspersed between the foregoing. For reliability samples see 4.5.4.1.4.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Finish, Protective	3.5
Marking	3.7
Semiconductors (Transistors & Diodes)	3.10

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to burn-in/debugging (See 3.16) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of burn-in/debugging shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government -- at its discretion -- may perform all or any part of the specified inspection. to verify the contractor's compliance with specified requirement. (See 6.6) Test equipmnet for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirement of Table I. Each lot will be subject to verification utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table I. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection 4.14 shall be last (See 6.4).

TABLE I

GROUP A INSPECTION

Inspection	Req. Para	Insp. Para	AQL Major %	AQL Minor %
Visual & Mechanical	3.20	4.12		
Modules (Pluck Out):				
VCO Circuit Card			1.0	4.0
Phase Detector Circuit Card			1.0	4.0
Fixed Divider Circuit Card			1.0	4.0
Control Divider Circuit Card			1.0	4.0
Lockout Logic Circuit Card			1.0	4.0
Encoder #2 Circuit Card			1.0	4.0
Encoder #1 Circuit Card			1.0	4.0
Servo Logic Circuit Card			1.0	4.0
Phase and Power Detector Circuit Card			1.0	4.0
Logic Circuit Card			1.0	4.0
RF Power Circuit Card			1.5	6.5
+22V Power Supply Circuit Card			1.0	4.0
Assemblies:				
Power Supply Circuit Card Assembly			1.5	6.5
Power Amplifier Circuit Card Assembly			1.0	4.0
Switch Filter Assembly			1.0	4.0
Motor Drive Assembly			1.0	4.0
Coupler Assembly			1.5	6.5
Electronic Component Assembly			1.0	4.0
Switch Filter Assembly			1.0	4.0
+8V Power Supply Assembly			1.0	4.0
Power Amplifier/Power Supply Assembly			1.5	6.5
Transmitter, Radio T-1199( )/TRN-30 (Assembled)			4.0	15.0
Amplifier-Coupler, AM-6417( )/TRN-30 (Assembled)			4.0	15.0
Cable, Audio			1.0	1.5
Cable, Battery			1.0	1.5

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TABLE I

GROUP A INSPECTION

Continued

Inspection	Req. Para	Insp. Para	AQL Major %	AQL Minor %
Antenna AS-2633( )/TRN-30			4.0	15.0
Antenna AS-2634( )/TRN-30			4.0	15.0
Case, Amplifier-Coupler CY-7143( )/TRN-30			1.5	6.5
Case, Transmitter CY-7142( )/TRN-30			1.5	6.5
Reel Assembly (u/w Transmitter)			1.0	1.5
Reel Assembly (u/w Amplifier-Coupler)			1.0	1.5
Electrical				
Transmitter, T-1199( )/TRN-30:				
Transmitter, lockout	3.14.2	4.10.2		
Exciter, lockout	3.14.4	4.10.4		
Frequency stability	3.14.5	4.10.5		
Code characters	3.14.6.1	4.10.6.1		
Code rate	3.14.6.2	4.10.6.2		
Word spacing	3.14.6.3	4.10.6.3	*1.0% for the combined group (TRANS & AMP-COUP)	
Percent modulation	3.14.6.5	4.10.6.5		
Power, steady state	3.14.7.1	4.10.7.1		
Antenna tuning	3.14.8	4.10.8		
Transmitter, RF	3.14.9.1	4.10.9.1		
Exciter, RF	3.14.9.2.1	4.10.9.2.1		
Exciter, audio	3.14.9.2.2	4.10.9.2.2		
Exciter, logic	3.14.9.2.3	4.10.9.2.3		
Meter, voltage	3.14.10	4.10.10		
Meter, RF	3.14.11	4.11.11		
Amplifier-Coupler AM-6417( )/TRN-30:				
Power, steady state	3.15.1.1	4.11.1.1		
Percent modulation	3.15.2	4.11.2		
Antenna tuning	3.15.3	4.11.3		
Tactical mode, RF	3.15.4.1	4.11.4.1		
Semi-fixed mode, RF	3.15.4.2	4.11.4.2		
Exciter enable	3.15.5	4.11.5		
Meter, RF	3.15.6	4.11.6		
Operational Inspection	3.18	4.14	0.1%	

*All electrical defects are considered major.

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.3 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II

GROUP B INSPECTIONS

Inspection	Req Para.	Insp. Para	AQL
Transmitter, T-1199( )/TRN-30:			
Transmitter, operating	3.14.1	4.10.1	
Exciter, operating	3.14.3	4.10.3	
Modulation frequency	3.14.6.4	4.10.6.4	6.5% for the
Overvoltage transient	3.14.7.2	4.10.7.2	Combined
Undervoltage transient	3.14.7.3	4.10.7.3	Group
Power consumption	3.14.7.4	4.10.7.4	
Amplifier-Coupler AM-6417( )/TRN-30:			
Overvoltage transient	3.15.1.2	4.11.1.2	
Undervoltage transient	3.15.1.3	4.11.1.3	
Power consumption	3.15.1.4	4.11.1.4	
Interchangeability (Mechanical)	3.17	4.13	6.5%
Interchangeability (Electrical)	3.17	4.13	6.5%

4.5.4 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

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4.5.4 Group C inspection (Cont'd)

TABLE III

GROUP C INSPECTION

Inspection	Req. Para	Insp. Para
Sub-Group I:		
Electromagnetic compatibility	3.13	4.9
Temperature	3.12.1	4.8.1
Altitude	3.12.3	4.8.3
Humidity	3.12.2	4.8.2
Sub-Group II:		
Dust	3.12.6	4.8.6
Rain	3.12.8	4.8.8
Salt Fog	3.12.9	4.8.9
Sub-Group III:		
Resonance search	3.12.4.1	4.8.4.1
Bench handling	3.12.5.1	4.8.5.1
Bounce (loose cargo)	3.12.4.2	4.8.4.2
Drop	3.12.5.2	4.8.5.2
Air delivery	3.12.11	4.8.11
Immersion	3.12.7	4.8.7
Fungus	3.12.10	4.8.10
Sub-Group IV:		
Reliability	3.12.12	4.8.12

4.5.4.1 Sampling for inspection. - Units shall be selected for each group of inspection in Table III as specified in 4.5.4.1.1 through 4.5.4.1.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s). Selection of samples shall be as specified and within 45 days (within 60 calendar days for Sub-Group IV) after selection of samples shall have been subjected to and completed the Group C Inspections. (See 6.3)

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4.5.4.1.1 Sub-Group I. - For this sub-group, three units from the first fifteen production units shall be selected for the first Group C inspection. For subsequent Group C Inspections, three units per month shall be selected. For the EMC inspection during production, methods CE 03, CS 06 and RE 02 shall be performed on one unit.

4.5.4.1.2 Sub-Group II. - For this sub-group, two units (these units may be the units selected for Sub-Group I) from the first fifteen production units shall be selected for the first Group C Inspection. For subsequent Group C Inspection, two units every two months shall selected.

4.5.4.1.3 Sub-Group III. - For this sub-group, two units from the first fifteen production units shall be selected for the first Group C Inspection. For subsequent Group C Inspection, two units shall be selected when one-half of the units on order have been accepted by the Government QA Representative.

4.5.4.1.4 Sub-Group IV. - For this sub-group, a minimum sample of five Transmitter units and five Transmitter/ Amplifier-Coupler units shall be selected for the Reliability Qualification Test (4.8.12.1) from the first article samples. For subsequent Group C inspections, a minimum sample of three Transmitter units and three Transmitter/Amplifier-Coupler units for the Reliability Acceptance Tests (4.8.12.2) shall be selected from every two months production.

4.5.5 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Product Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, the units are cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

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4.6 Burn-in/debugging. - Each Transmitter T-1199( )/TRN-30 and each Amplifier-Coupler AM-6417( )/TRN-30 submitted for acceptance shall be subjected to Burn-in in accordance with Drawing SC-A-678026. The requirements of 3.16 shall be met.

4.7 Standard Test Conditions. - Unless otherwise specified, inspections shall be performed at room temperature of 73 ± 18°F and humidity not to exceed 80% under the following conditions:

Transmitter, T-1199( )/TRN-30

a. Input Power: 28.0 ± 0.5 volts dc

b. Code Select Switches: ANAN

c. Code Rate Control: Full CCW

d. Mode Select Switch: MCW

e. Antenna Load: A series capacitive/resistive load shall be connected between the antenna connector, J4 and a ground radial connector. For each operating frequency, the characteristics of this load shall be in accordance with the following table: (NOTE: When this antenna load is used during EMC testing, it shall be shielded and mounted directly on top of the RF output terminal of the unit under test)

TRANSMITTER T-1199( )/TRN-30(V)

ANTENNA LOAD

Frequency (kHz)	Antenna Switch (feet)	Resistance (±5%) (ohms)	Capacitance (±2%) (pf)
200.0 - 259.5	30	7.5	290
260.0 - 309.5	30	6.5	290
310.0 - 359.5	30	5.5	290
360.0 - 535.5	30	4.7	290
1605.0 - 1750.5	30	3.2	340
1605.0	15	7.5	150
1750.0	15	11.0	670

f. Warm-up Time: A maximum of 30 seconds after completion of tuning.

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Amplifier-Coupler AM-6417( )/TRN-30

a. Input Power: 28.0 ± 0.5 volts dc

b. RF Drive (J3) square wave at the operating frequency with the following characteristics:

(1) Amplitude: 31 ± 1.0 volts peak to peak

(2) DC Offset: Zero volts dc

(3) Rise Time: Less than 50 ± 20 nano seconds

(4) Symmetry: Half cycle zero-crossings symmetrical within 1%.

(5) Power: 4.0 ± 1.0 watts

c. Filter Select Signals: Filter select signals shall conform to the following table.

FILTER SELECT SIGNALS

	CONNECTOR J2 (VDC)			
Frequency (kHz)	Pin C	Pin P	Pin N	Pin L
200.0 - 259.5	4.5±0.5	0.0	0.0	0.0
260.0 - 309.5	0.0	4.5±0.5	0.0	0.0
310.0 - 359.5	0.0	0.0	4.5±0.5	0.0
360.0 - 535.0	0.0	0.0	0.0	4.5±0.5

d. Audio Input: A balanced audio signal shall be connected to connector J2 pins L and N with the return to Pin F and shall have the following characteristics:

(1) Amplitude: 0.70 ± 0.05 volts RMS

(2) Frequency: 1020 ± 10 Hz

(3) Duty Cycle: 50% maximum with on time not to exceed five seconds.

e. Antenna Load: A series capacitive/resistive load shall be connected between the antenna connector and a ground radial connector. For each operating frequency, the characteristics of this load shall be in accordance with the following table. (NOTE: When this antenna load is used during EMC testing, it shall be shielded and mounted directly on top of the RF output terminal of the unit under test.)

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AMPLIFIER-COUPLER AM-6417( )/TRN-30

ANTENNA LOAD

Frequency (kHz)	Resistance (±5%) (ohms)	Capacitance (±2%) (pf)
200.0 - 259.5	8.2	500
260.0 - 309.5	7.1	500
310.0 - 359.5	6.0	500
360.0 - 535.0	5.5	500

f. Unless otherwise specified, all measurements shall be made in the semi-fixed modes.

4.7.1 Standard Test Frequencies. - Unless otherwise specified, the following standard test frequencies shall be used:

4.7.1.1 Transmitter T-1199( )/TRN-30. - 200.0 kHz, 227.0 kHz, 259.5 kHz, 260.0 kHz, 309.5 kHz, 310.0 kHz, 359.5 kHz, 360.5 kHz, 535.5 kHz, 1605.0 kHz, and 1750.0 kHz.

4.7.1.2 Amplifier-Coupler AM-6417( )/TRN-30. - 200.0 kHz, 227.0 kHz, 259.5 kHz, 260.0 kHz, 309.5 kHz, 310.0 kHz, 359.5 kHz, 360.0 kHz, and 535.0 kHz.

4.8 Service conditions. - Service Condition Tests shall be conducted to meet the requirements of 3.12. Units shall meet the tests as specified in Table IV. No defects will be allowed.

4.8.1 Temperature. -

4.8.1.1 High temperature. - The equipment shall be subjected to the test of Method 501, Procedure II of MIL-STD-810. The high operating temperature for the equipment shall be 155° F. (This includes effect of sun load). This test shall be performed to determine compliance with 3.12.1.1.

4.8.1.2 Low temperature. - The equipment shall be subjected to the test of Method 502, Procedure I of MIL-STD-810. The storage temperature shall be -65°F and the lowest operating temperature shall be -50°F. This test shall be performed to determine compliance with 3.12.1.2.

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TABLE IV

		Low Temp		High Temp		Rain		Humi-dity		Air Del		Alt.		Dust		Vib		Shock	
Req. Para	Freq kHz	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During	Before/After	During
TRANSMITTER																			
3.14.1	200 280	X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.5	335 535	X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.6.3	1605 1750	X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.6.5		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.7.4		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.8		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.9.1		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.14.11		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
AMPLIFIER-COUPLER																			
3.15.1.4	200 280	X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.15.2	335 535	X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
3.15.4.2		X	X	X	X	X	X	X	X	X	NA	X	X	X	NA	X	NA	X	NA
Mechanical & Visual Inspection and also Operation of Controls (See 3.4)		To be conducted before and after each Service Conditions Test.																	

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4.8.2 Humidity. - The equipment shall be subjected to the test of Method 507, Procedure III of MIL-STD-810. Measurements shall be made during the last 5 hours of each cycle of step 6. This test shall be performed to determine compliance with 3.12.2.

4.8.3 Altitude. - The equipment shall be subjected to the test of Method 500, Procedure I of MIL-STD-810. This test shall be performed to determine compliance with 3.12.3.

4.8.4 Vibration. -

4.8.4.1 Resonance search. - The equipment shall be subjected to the test of Method 514, Procedure IX, Part 2 of MIL-STD-810. This test shall be performed to determine compliance with 3.12.4.1

4.8.4.2 Bounce, (loose cargo). - The equipment shall be subjected to the test of Method 514, Procedure XI, Part 1 of MIL-STD-810. This test shall be performed to determine compliance with 3.12.4.2.

4.8.5 Shock. -

4.8.5.1 Bench handling. - The equipment shall be subjected to the test of Method 516, Procedure V of MIL-STD-810. This test shall be performed to determine compliance with 3.12.5.1.

4.8.5.2 Drop. - The equipment shall be subjected to the test of Method 516, Procedure II of MIL-STD-810. This test shall be performed to determine compliance with 3.12.5.2.

4.8.6 Dust. - The equipment shall be subjected to the test of Method 510, Procedure I of MIL-STD-810. This test shall be performed to determine compliance with 3.12.6.

4.8.7 Immersion. - The equipment in its transit cases shall be subjected to the test of Method 512, Procedure I of MIL-STD-810. The transit cases shall be in a closed condition during this test. This test shall be performed to determine compliance with 3.12.7.

4.8.8 Rain. - The equipment shall be subjected to the test of Method 506, Procedure I of MIL-STD-810. This test shall be performed to determine compliance with 3.12.8.

4.8.9 Salt Fog. - The equipment shall be subjected to the test of Method 509, Procedure I, MIL-STD-810. Electrical operation of the equipment shall not be required. There shall be no evidence of corrosion after completion of the test. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. This test shall be performed to determine compliance with 3.12.9.

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4.8.10 Fungus. - The equipment shall be subjected to the test of Method 508, Procedure I of MIL-STD-810. All gasket sealed or other non-hermetic sealed equipments shall be sprayed with spore suspension and exposed with the chassis removed and open from the case. Operation of the equipment shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus on any surfaces. This test shall be performed to determine compliance with 3.12.10.

4.8.11 Air delivery. - The equipment shall be subjected to a simulated air drop impact test in accordance with MIL-STD-669. This test shall be performed to determine compliance with 3.12.11.

4.8.12 Reliability (See 3.12.12). -

4.8.12.1 Reliability Qualification Test. - The reliability of the Transmitter T-1199( )/TRN-30 shall be demonstrated in accordance with Test Plan IV of MIL-STD-781 using a specified MTBF (θ_0) of 1200 hours. The reliability of the Transmitter T-1199( )/TRN-30 and Amplifier-Coupler AM-6417( )/TRN-30 shall be demonstrated by testing only the semi-fixed mode in accordance with Test Plan IV of MIL-STD-781, using a specified MTBF (θ_0) of 1000 hours. The reliability testing shall be conducted as specified in MIL-STD-781.

4.8.12.2 Reliability Acceptance Test. - The reliability of the equipments comprising production lots shall be demonstrated in accordance with Test Plan IVA of MIL-STD-781, using a specified MTBF (θ_0) of 1200 hours for the Transmitter T-1199( )/TRN-30 and a specified MTBF (θ_0) of 1000 hours for the Transmitter T-1199( )/TRN-30 and Amplifier-Coupler AM-6417( )/TRN-30 in the semi-fixed mode. The reliability testing procedures shall be those approved during reliability qualification testing. Reliability testing will be performed only on samples which have successfully completed burn-in and have passed Group A and B inspection.

4.8.12.3 Reliability Test Conditions. - All reliability testing shall be conducted in accordance with Test Level C of MIL-STD-781. Proper instrumentation shall be provided to insure immediate recognition of failure. A failure is defined as the inability of an equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure from performance as required by this specification and the contractor's approved test procedure.

4.8.12.4 Rework and Retest Provision. - Once the lot has been rejected, no rerun can be made of that test on the same or a new sample from the same lot unless positive action consisting of design change, modification or rework actually related to the observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time. Repair of the failed equipment will be limited to replacement of the defective assembly, sub-assembly, card or part, with an identical burned-in item. The failed item will then undergo analysis to determine the cause of failure.

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4.8.12.5 Test Length Per Equipment. - No single equipment on time shall be less than one half the average operating time of all equipments on test.

4.8.12.6 Required daily measurements. - Performance parameters to be monitored for each equipment during the reliability tests are as follows:

Transmitter: 3.14.1, 3.14.8, 3.14.6.1, 3.14.6.2, 3.14.6.3, 3.14.6.4 and 3.14.9.1. Each of these parameters shall be performed at 200.0, 300.0, 400.0, 500.0, and 1700.0 kHz.

Transmitter and Amplifier-Coupler: 3.14.6.1, 3.14.6.2, 3.14.6.3, 3.14.6.4 (NOTE: These parameters are performed on the exciter), 3.15.2, 3.15.4.2 and 3.15.3. Each of these parameters shall be performed at 200.0, 300.0, 400.0 and 500.0 kHz.

4.9 Electromagnetic compatibility. - The Transmitter/Amplifier-Coupler(s) shall be tested for compliance with 3.13. The test setups and test procedures shall be in accordance with the measurement techniques of MIL-STD-462 except as indicated herein.

4.9.1 Conducted emission CE 01. - This test shall be performed to determine compliance with 3.13.1.

4.9.2 Radiated emission RE 02. - Place the measurement antenna 65 feet from the base of the equipment being tested. Measurements shall be performed to determine compliance with 3.13.2.

4.9.3 Spurious and harmonic emissions RE 03. - Utilizing one frequency in each band, under standard test conditions, the fundamental power output of the equipments in all modes shall be measured simulating the antenna impedance at the fundamental operating frequency. Remove simulated loads and connect applicable antenna. The field measurement antenna shall be placed at a 120 foot distance. The fundamental power output shall be measured in terms of field strength (dB/uV/m). Upon completion of the fundamental field strength measurement, the fundamental frequency shall be rejected and the required field scanned for harmonic and spurious emissions. Their field strength values shall then be subtracted from the fundamental measurement to calculate the dB suppression. This technique eliminates the necessity of compensating for unknown antenna gain parameters. These measurements shall be performed to determine compliance with 3.13.3.

4.9.4 Radiated susceptibility RS 03. - This test shall be performed to determine compliance with 3.13.4.

4.10 Electrical requirements, Transmitter T-1199( )/TRN-30. - All measurements are to be performed under standard conditions, unless otherwise noted.

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4.10.1 Transmitter, operating. - Set the mode switch to the CW position, and measure the RF frequency at each frequency in 4.7.1.1 at 22, 28 and 30 ±0.5 VDC. These measurements shall be performed to determine compliance with 3.14.1.

4.10.2 Transmitter, lockout. - With the DC input current less than 4 amps and the battery meter in the green area, check for zero RF and frequency indications at 199.5, 536.0, 1604.0 and 1751.0 kHz to determine compliance with 3.14.2.

4.10.3 Exciter, operating. - Set the mode switch to the CW position and measure the RF frequency at each frequency in 4.7.1.2 at 26, 28 and 30 ±0.5 VDC. These measurements shall be performed at the RF output connector to determine compliance with 3.14.3.

4.10.4 Exciter, lockout. - With the DC input current less then 30 amps and the battery meter in the green area, check for zero RF and frequency indications at 199.5, 536.0, 1604.0, 1604.5, 1750.0, and 1750.5 kHz. These measurements shall be performed at the RF output connector to determine compliance with 3.14.4.

4.10.5 Frequency stability. - Set the mode switch to the CW position and measure the output frequency at each frequency in 4.7.1.1 at 22, 28 and 30 ±0.5 VDC. These measurements shall be performed to determine compliance with 3.14.5.

4.10.6 Beacon identification. -

4.10.6.1 Code characters. - Operate the transmitter at 200.0 kHz in the MCW mode with the code word "AAAA." Observe that the RF carrier modulation consists of the Morse Code equivalent of the code word. Repeat using the remaining 25 letters of the alphabet and blank to determine compliance with 3.14.6.1.

4.10.6.2 Code rate. - Operate the transmitter at 200.0 kHz in the MCW mode with the code word "HHHH." Observe the dot length of the modulated RF carrier. Turn the Code Rate Control full CCW and determine that the dot length is greater than 150 milliseconds. Turn the Code Rate Control full CW and determine that the dot length is less than 50 milliseconds. Determine that the dash length is three times that of the dot length with word "AAAA." These measurements shall be performed to determine compliance with 3.14.6.2.

4.10.6.3 Word spacing. - Operate the transmitter at 200.0 kHz in the MCW mode with Code Word "EEEE." Observe the modulated RF carrier and determine that the time betwen successive words is 3.5 ± 1 second. This measurement shall be performed to determine compliance with 3.14.6.3.

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4.10.6.4 Modulation frequency. - Operate the transmitter at 200.0, 260.0, 310.0, 360.0, 535.0, and 1605.0 kHz in the MCW mode with the code word "0000" and measure the frequency of the amplitude modulation to determine compliance with 3.14.6.4 at each frequency.

4.10.6.5 Percent modulation. - Operate the transmitter in the MCW mode with code word "0000" and measure the percent modulation at each frequency to determine compliance with 3.14.6.5.

4.10.7 Input power. -

4.10.7.1 Power, steady state. - The equipment shall meet full specification performance when operated under standard conditions. Adjust the input voltage to 22.0 VDC and 30.0 VDC. Measure the RF antenna current and frequency at 200 kHz to determine compliance with 3.14.7.1.

4.10.7.2 Overvoltage transient. - Under standard conditions at 200.0 kHz apply a 40 volt pulse between 0.1 second and 0.5 second duration to the power input. The requirements of 3.14.7.2 shall be met.

4.10.7.3 Undervoltage transient. - Under standard conditions at 200.0 kHz decrease the input voltage to 15 volts for one second. The requirements of 3.14.7.3 shall be met.

4.10.7.4 Power consumption. - Under standard conditions, measure the input current to determine compliance with 3.14.7.4.

4.10.8 Antenna tuning. - Using a timing device, observe the time required to tune the transmitter after the power switch is turned on. Tuning is completed when the code is audible. Record the tuning time to determine compliance with 3.14.8.

4.10.9 Output signals. -

4.10.9.1 Transmitter, RF. - Set the mode switch to the CW position and measure the RF current at each frequency in 4.7.1.1 at 22, 28 and 30 +0.5VDC. These measurements shall be performed to determine compliance with 3.14.9.1.

4.10.9.2 Exciter. -

4.10.9.2.1 Exciter, RF. - Set the mode switch to the CW position and measure the RF output power at 200.0, 285.0, 325.0 and 535.0 kHz at 26, 28 and 30 +0.5 VDC. These measurements shall be performed at the RF output connector to determine compliance with 3.14.9.2.1.

4.10.9.2.2 Exciter, audio. - Switch the mode switch to KEY and measure the audio output power and frequency at 200.0 kHz to determine compliance with 3.14.9.2.2.

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4.10.9.2.3 Exciter, logic. - Set the mode switch to the CW position and measure the logic signal levels at the audio connector at 200.0, 260.0, 310.0 and 360.0 kHz to determine compliance with 3.14.9.2.3.

4.10.10 Meter, voltage. - Under standard conditions at 200.0 kHz, operate the equipment at an input voltage of 22.0 volts and determine that the meter is indicating the lower edge of the green zone. Increase the input voltage to 30.0 volts and determine that the meter is indicating the upper edge of the green zone. The requirements of 3.14.10 shall be met.

4.10.11 Meter, RF. - Under standard conditions, determine that the meter is indicating in the green area to determine compliance with 3.14.11.

4.11 Electrical requirements, Amplifier-Coupler AM-6417( )/TRN-30. - All measurements are to be performed under standard conditions, unless otherwise noted.

4.11.1 Input power. -

4.11.1.1 Power, steady state, - The equipment shall meet full specification performance when operated under standard conditions. Adjust the input voltage to 26.0 and 30.0 VDC. Measure the RF antenna current and frequency at 200.0 kHz to determine compliance with 3.15.1.1.

4.11.1.2 Overvoltage transient. - Under standard conditions at 200.0 kHz apply a 40 volt pulse between 0.1 second and 0.5 second duration to the power input. The requirements of 3.15.1.2 shall be met.

4.11.1.3 Undervoltage transient. - Under standard conditions at 200.0 kHz, decrease the input voltage to 15 volts for one second. The requirements of 3.15.1.3 shall be met.

4.11.1.4 Power consumption. - Under standard conditions, measure the input current to determine compliance with 3.15.1.4.

4.11.2 Percent modulation. - Under standard test conditions, apply the modulated signal in 4.7 with the Code Word "0000" and measure the percent modulation at each frequency to determine compliance with 3.15.2.

4.11.3 Antenna tuning. - Using a timing device, observe the time required to tune the Amplifier-Coupler after the power switch is turned on. Tuning is completed when the code is audible. Record the tuning time to determine compliance with 3.15.3.

4.11.4 Output power. -

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4.11.4.1 Tactical mode, RF. - Apply a CW signal and operate the Amplifier-Coupler in the tactical mode. Measure the RF current at each frequency in 4.7.1.2 at 26, 28 and 30 ±0.5VDC to determine compliance with 3.15.4.1.

4.11.4.2 Semi-fixed mode, RF. - Apply a CW signal and operate the Amplifier-Coupler in the semi-fixed mode. Measure the RF current at each frequency in 4.7.1.2 at 26, 28 and 30 ±0.5 VDC to determine compliance with 3.15.4.2.

4.11.5 Exciter enable. - Connect the audio cable to the transmitter and amplifier-coupler. Apply the required input power. Determine that the RF power and the frequencies of 1605 and 1750 kHz are not present at the transmitter antenna terminal. Measure the required RF output at 200.0 and 535.0 kHz at the RF connector. These measurements shall be performed to determine compliance with 3.15.5.

4.11.6 Meter, RF. - Under standard conditions, determine that the meter is indicating in the green area to determine compliance with 3.15.6.

4.12 Visual and mechanical inspection. - Assembled unit(s) shall be examined for defects listed in MIL-STD-252. Prior to assembly into the unit(s), printed circuits boards shall be examined for compliance with MIL-P-55110 and the equipment drawings. Assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-252. The requirements of 3.20 shall be met.

4.13 Interchangeability. - The mechanical dimensions and electrical performance of parts will be physically and electrically interchangeable between the unit(s). Mechanical dimensions and electrical performance required by the gage and equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.17. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions and performance shall be gaged or measured prior to assembly into the units:

a. Mechanical (dimensions). - The gage points covered by the drawings listed on the applicable gage lists specified in the solicitation/ contract.

b. Electrical performance (Items). - Items shall be tested to determine compliance with the performance requirements specified on the applicable gage lists specified in the solicitation/contract.

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4.14 Operational inspection. - Each unit(s) shall be operated under standard conditions and standard frequencies in accordance with 4.7. The units shall provide normal RF antenna current, percent modulation, frequency and code word. The requirements of 3.18 shall be met.

4.15 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance with the packaging requirements of Section 5.

4.16 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the operational inspection test 4.14 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Beacon Set, Radio AN/TRN-30( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Beacon Set, Radio AN/TRN-30( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation applications. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Beacon Set, Radio AN/TRN-30( ). - Each Beacon Set, Radio AN/TRN-30( ) shall be individually packaged method III as follows: Stow the units of each set within the designated compartments of the transit cases CY-7142 or CY-7143. Close case cover, and secure with fasteners provided. Fabricate corner pads of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5C. Secure the corner pads to each case with tape conforming to PPP-T-76. Place one technical literature, packaged as specified in 5.1.1.4.1 directly on top of each case and secure with tape conforming to PPP-T-76. No further packaging required.

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5.1.2 Level C. - Beacon Set, Radio AN/TRN-30( ) in transit cases CY-7142 or CY-7143 shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2)

5.2.1 Level A. - Beacon Set, Radio AN/TRN-30( ) in transit cases CY-7142 or CY-7143, packaged as specified in 5.1 shall be individually packed within a close-fitting box conforming to PPP-B-601 overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. Skids, 3x4 inch, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. - Beacon Set, Radio AN/TRN-30( ) in transit cases CY-7142 or CY-7143 packaged as specified in 5.1 shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. Skids, 3x4 inch, laid flat shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or Appendix thereto.

5.2.3 Level C. - Beacon Set, Radio AN/TRN-30( ) in cases CY-7142 or CY-7143 packaged as specified in 5.1 shall be individually packed within shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage during limited shipment, handling and intransit storage.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Beacon Set, Radio AN/TRN-30( ) transmits an omnidirectional signal which is used in conjunction with airborne direction finding (ADF) radio receivers as a navigation aid. This is accomplished

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by two different configurations. In the Pathfinder configuration, Beacon Set, Radio AN/TRN-30( )(V)1 is used as a low power, short range, manportable direction finding beacon. In the Pathfinder configuration, the transmitter will operate with battery BA-525( )/U at temperatures down to 0°F or with an external power supply down to -50°F. In the Tactical and Semi-Fixed configuration, Beacon Set, Radio AN/TRN-30( )(V)2 is set up in a semi-permanent installation and used as a medium-to-high power and medium-to-long range direction finding beacon which is powered by an external power supply.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing required. (See Section 5)

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection - These sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.2)

e. When rough handling and functional tests are required.

f. Marking and shipping of sample.

g. Place of final inspection.

h. Technical literature required.

i. Quantity of spare parts required.

j. Submission of the statement of treatment referenced in 3.11.4 should be as specified in the solicitation/contract.

6.3 Group C inspection. - Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.4 is complete.

6.4 Location of operational inspection. - It is desirable that the operational inspection 4.14 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

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6.5 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TRN-30W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and quality history of the product.

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